NUS GRIP
Ambient Systems
MVP Budget Request
Version: 01
March 4, 2021

Intro & Background

Ambient is an NUS GRIP startup with a mission to empower building operators to easily manage the indoor climate. Most solutions in this space are focused on optimizing a single goal, such as reducing energy costs. Ambient's product consists of an end-to-end hardware stack, dashboard, and optimization model leverage latest innovations in machine learning to manage multiple critical yet competing priorities. In particular, the goal of the MVP is to demonstrate the optimization capabilities that a customer can see in their building.

Scope
We are seeking a budget to develop a portion of the software as part of the MVP. The goal is to use the MVP to gain further traction with customers and demonstrate the technology to the GRIP investment panel in May. The initial product will be focused on the indoor farming segment based on the UI designs in

The initial feature list based on the UI designs:
1. Display data in different chart formats (bar graph, scatter plot, radar chart, pie chart) (4-6 objects)
2. Map interface (indoor map of the floor plan with various equipment)
3. Filter (and sort) data by time (and other attributes)
4. Export data in json format
5. Basic user inputs (user can input and toggle preferred settings) (4-6 input settings)

Deliverables

A single-page web dashboard to display real-time data information and provide way for user to input their preferences into the system.

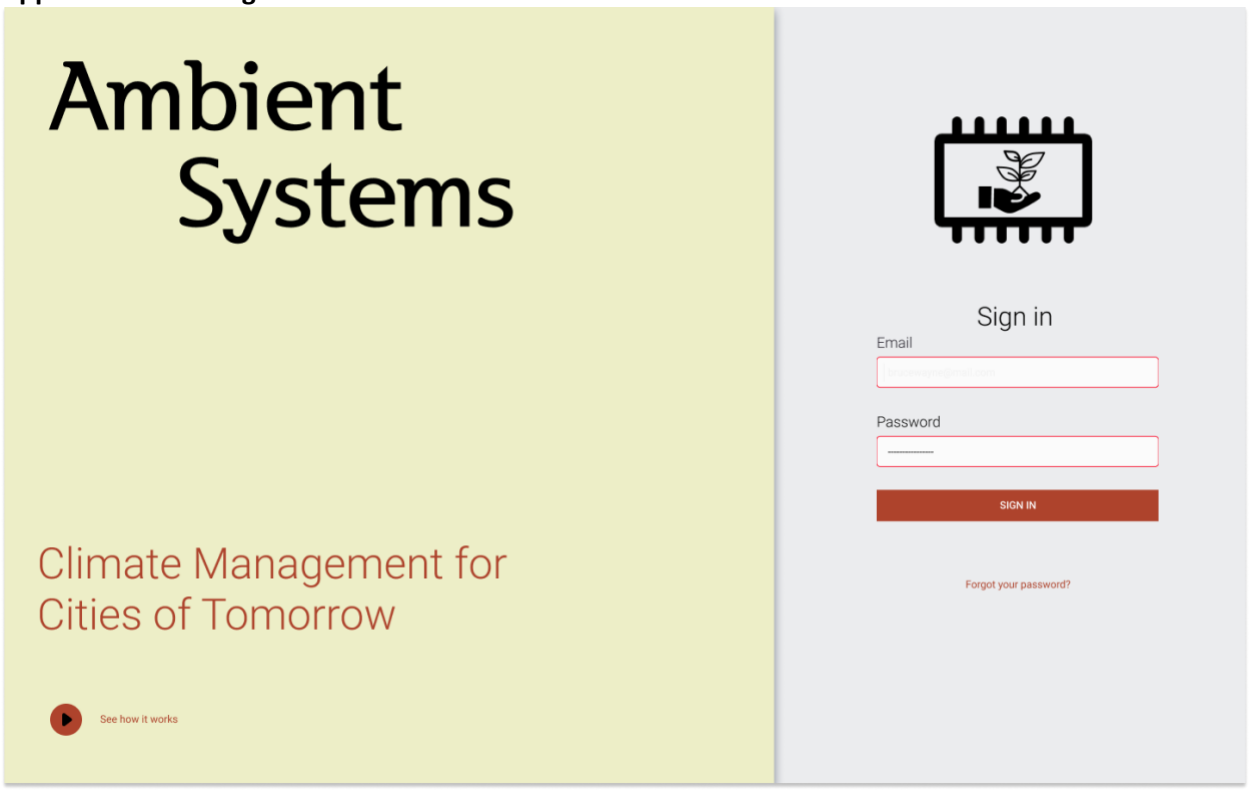
Schedule with Milestones

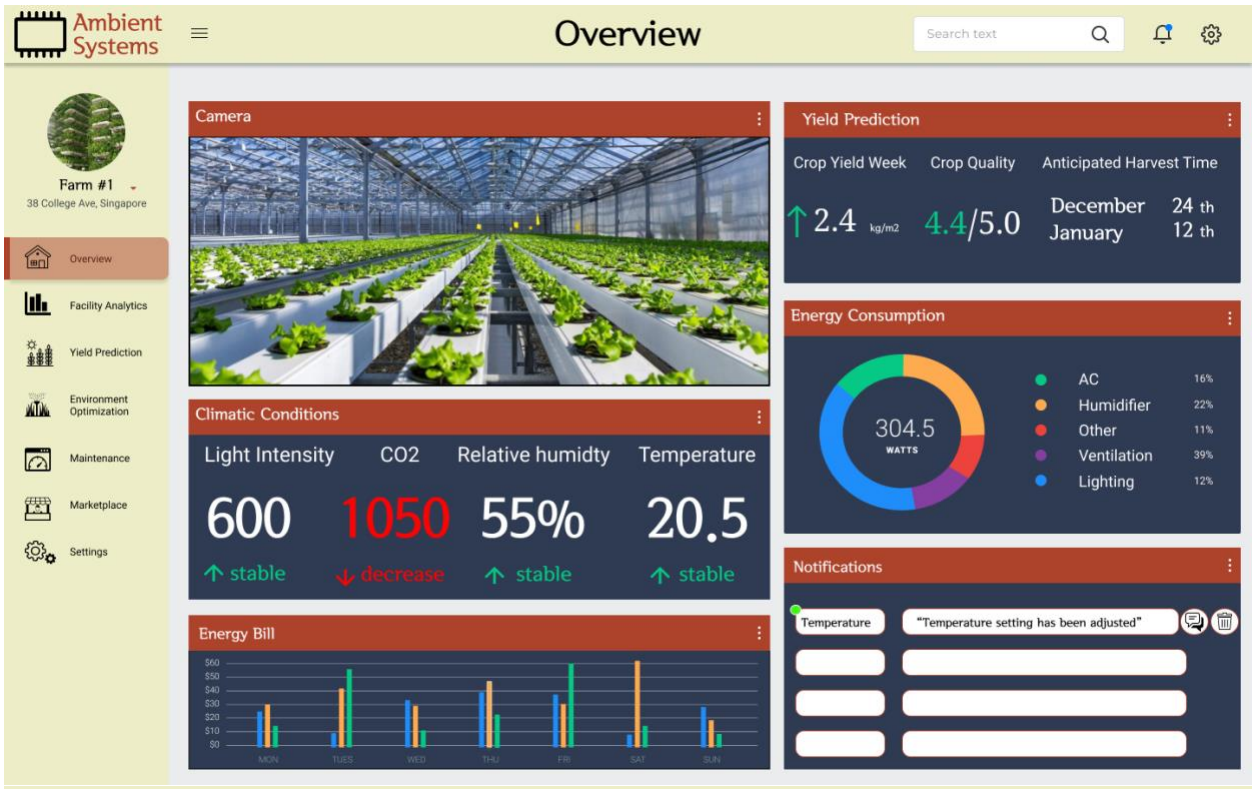
- Mar 10: tentative start
- Mar 17: v01 complete
    - Feature 1
    - Feature 3
- Apr 1: v02 complete
    - Feature 2
    - Feature 5
- May 14: v03 (final complete)
    - Feature 4

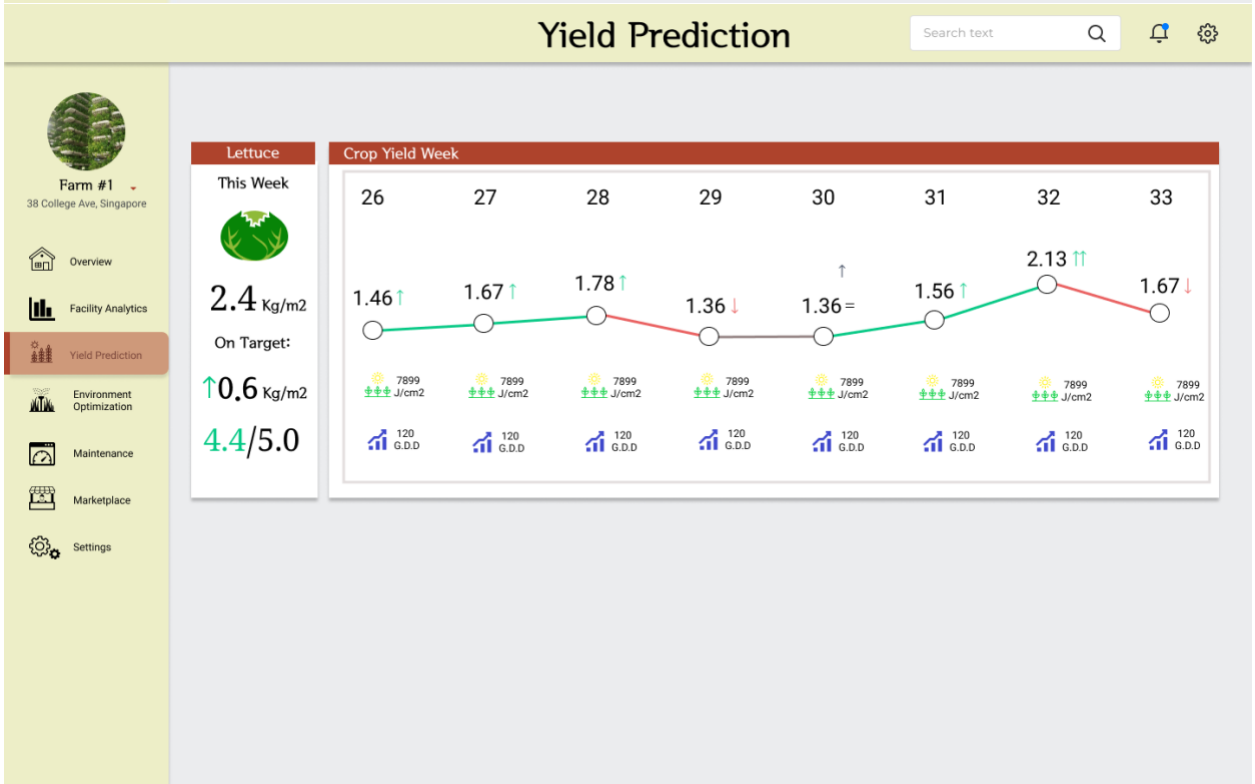
Cost

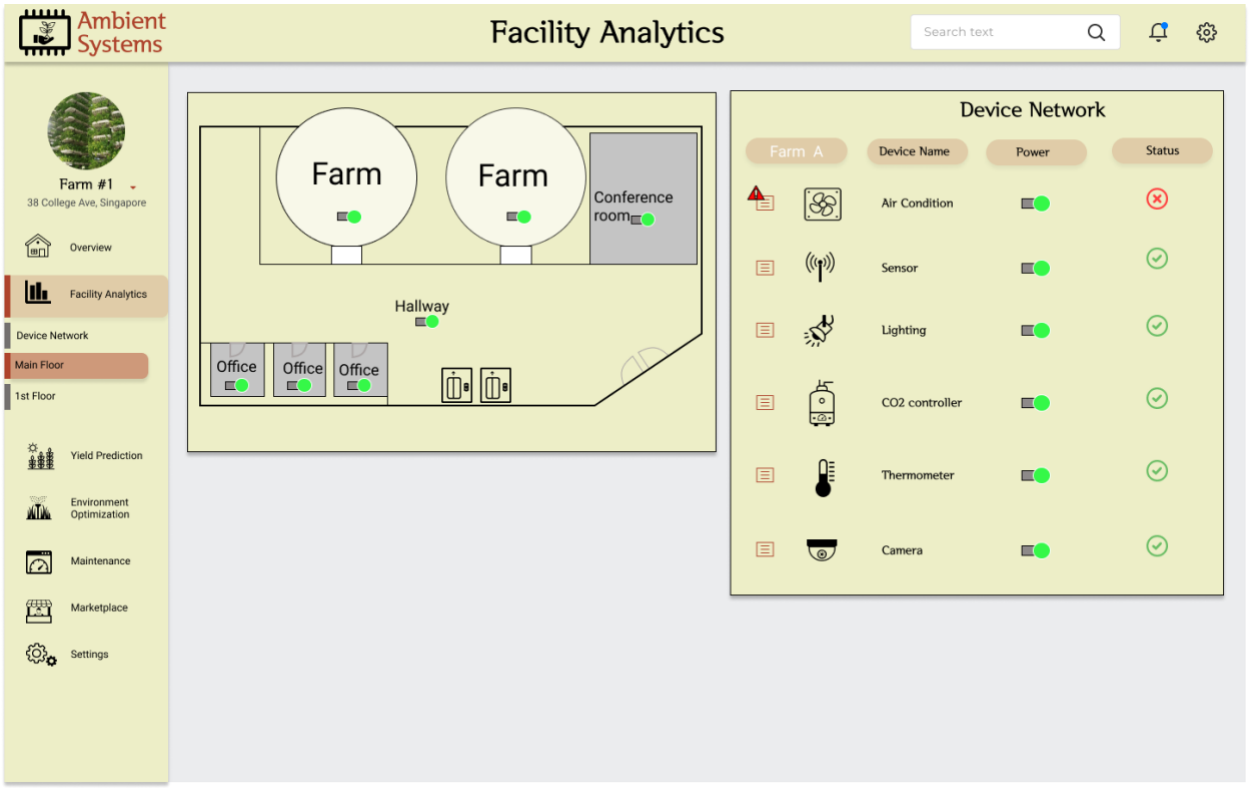
The estimated cost is based on the amount of hours that the developer will need to implement the MVP. The fidelity of the dashboard will impact the cost. The current cost estimate is between $3,000 and $6,000.

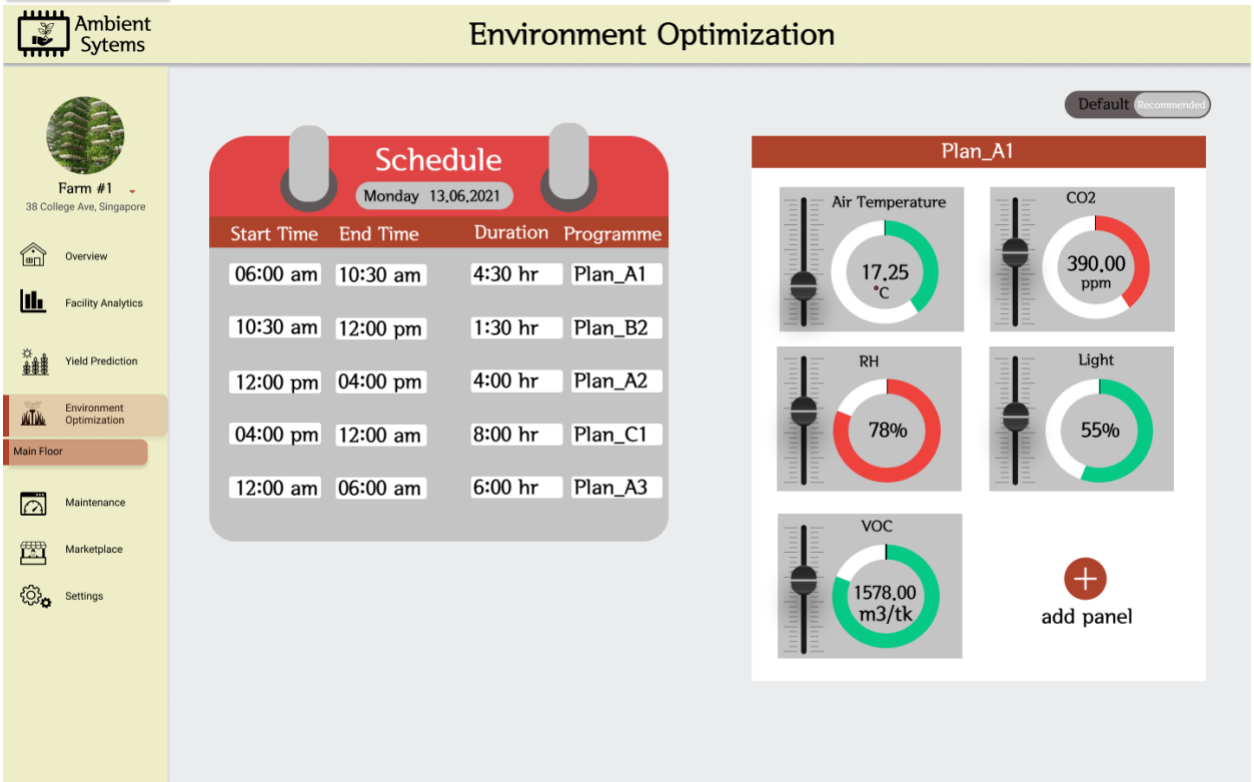
Appendix A: UI Design




Ambient Systems
Overview
Search text
Farm #1
38 College Ave, Singapore
Overview
Facility Analytics
Yield Prediction
Environment Optimization
Maintenance
Marketplace
Settings
Camera
Yield Prediction
Crop Yield Week
Crop Quality
Anticipated Harvest Time
2.4 kg/m2
4.4/5.0
December 24 th
January 12 th
Energy Consumption
304.5
WATTS
AC 16%
Humidifier 22%
Other 11%
Ventilation 39%
Lighting 12%
Climatic Conditions
Light Intensity
CO2
Relative humidty
Temperature
600
1050
55%
20.5
stable
decrease
stable
stable
Notifications
Temperature
"Temperature setting has been adjusted"
Energy Bill
$60
$50
$40
$30
$20
$10
$0
MON
TUES
WED
THU
FRI
SAT
SUN


Yield Prediction
Search text
Farm #1
38 College Ave, Singapore
Overview
Facility Analytics
Yield Prediction
Environment Optimization
Maintenance
Marketplace
Settings
Lettuce
This Week
2.4 Kg/m2
On Target:
0.6 Kg/m2
4.4/5.0
Crop Yield Week
26
27
28
29
30
31
32
33
1.46
1.67
1.78
1.36
1.36
1.56
2.13
1.67
7899 J/cm2
120 G.D.D


Ambient Systems
Facility Analytics
Search text
Farm #1
38 College Ave, Singapore
Overview
Facility Analytics
Device Network
Main Floor
1st Floor
Yield Prediction
Environment Optimization
Maintenance
Marketplace
Settings
Farm
Farm
Conference room
Hallway
Office
Office
Office
Device Network
Farm A
Device Name
Power
Status
Air Condition
Sensor
Lighting
CO2 controller
Thermometer
Camera


Ambient Sytems
Environment Optimization
Farm #1
38 College Ave, Singapore
Overview
Facility Analytics
Yield Prediction
Environment Optimization
Main Floor
Maintenance
Marketplace
Settings
Schedule
Monday 13,06,2021
Start Time
End Time
Duration
Programme
06:00 am
10:30 am
4:30 hr
Plan_A1
10:30 am
12:00 pm
1:30 hr
Plan_B2
12:00 pm
04:00 pm
4:00 hr
Plan_A2
04:00 pm
12:00 am
8:00 hr
Plan_C1
12:00 am
06:00 am
6:00 hr
Plan_A3
Default
Recommended
Plan_A1
Air Temperature
17.25 °C
CO2
390.00 ppm
RH
78%
Light
55%
VOC
1578.00 m3/tk
add panel

Appendix B: Quotes

From:
Subject: Toptal Call Summary from Michal
Date: March 2, 2021 at 4:42 PM
To:

MR

- External Email -

Hi Ivan,

Thank you for your time earlier, it was great speaking with you!

As a summary, Toptal is a **professional services partner** for organizations that need to get quality work done efficiently and who don't have the internal resources to do it.

Our freelance network has the **top 3% of talent in the industry in 5 key verticals**, namely Software Development, Design, Project Management, Product Management, and Finance.

Through our **Matching process**, our Directors would hand-select the best candidates specifically for you for each respective role based on your predefined requirements.

Here is a list of some of our clients, as well as some client reviews.

> **Our Rigorous Vetting Process** (more details here)
>
> - Testing candidates on communication, personality, and technical skills
> - Accept only 3% of candidates worldwide
>
> **Engagement Types - flexible structure**
>
> - Full-time: 40+ hours per week
> - Part-time: 20+ hours per week
> - Hourly: Less than 20-hour per week
>
> **Time and Materials Model - You pay for hours worked:**
> * Rates vary primarily on the location of the expert, not skillset.
> ** 20% Discount applies to Part-Time (20 hours a week) or Full-Time (40 hours a week) Rates
>
> - Developers ($60-$150+/hour): https://www.toptal.com/developers/all
> - Designers ($70-$150+/hour): https://www.toptal.com/designers/all
> - Project Managers ($90-$180+/hour): https://www.toptal.com/project-managers/all
> - Product Managers ($90-$200+/hour): https://www.toptal.com/product-managers/all
> - Finance Experts ($90-$250+/hour): https://www.toptal.com/finance/all
>
> **No-Risk Trial Period (per expert hired)**
>
> - Varies in length up to 1 week, depending on the scope of your project
> - Satisfaction guaranteed
> - Begins after you interview and choose an expert
>
> To start the first Matching Process, we simply ask for a **fully-refundable $500 security deposit.** This is not a fee -- it will either be credited to your first invoice (if you approve your No-Risk Trial) or refunded in full (if you don't).

Next steps:
If you wish to get started, we will go through a quick activation of your account over the phone, which won't take more than a few minutes. In short, we will go through the following few steps.

- Step 1: I create your account profile, add a job description(s) to it and approve that profile;
- Step 2: You create your password and verify your billing information by adding in a valid payment method;
- Step 3: We schedule your Matching call(s);

Please note that the $500 fully-refundable deposit is held only after your first Matching Call

Look forward to hearing back.

View all Proposals



Kelvin A.

Hamburg, Germany
12:09pm local time

Applicant
Kelvin has applied to or been invited to your or your company's job *Customize IoT Admin Platform/Dashboard*

Proposal Details

$2,500.00
Proposed Bid

Cover letter

What frameworks have you worked with?
I'm mostly working with react, django/flask but I've also worked with vue, angular, etc...

Hi!

I'm Kelvin from Hamburg, Germany.
Your Job posting sounds really interesting!

I have 12 Years+ experience in Data Analysis, Web Scraping, Automation, Software Development and Web Development.

Here is an excerpt of the technologies and concepts I'm using:
Python, Pandas, Django, Data Mining, Web Scraping, Selenium, BS4 (Beautiful Soup), numpy, scipy, plotly, scikit-learn, tensorflow, pytorch, Rust, Kotlin, SQL & NoSQL, SQLight, MySQL, PostgreSQL, MongoDB, influxdb, Interactive Brokers API, IB, Pine Script, Equilla, Easy Language, Tradesignal, Tradestation, HTML/CSS, PHP, JavaScript, Ajax, Wordpress, React, AWS, Kubernetes, Docker

I've done Data Analysis projects targeting Financial Markets which includes writing Indicators, standalone Trading Systems, Data Mining, Machine Learning and Predictive Analytics.

View all Proposals



Gurjeet S.

Ferozepur, India
4:35pm local time

99%
Job Success

TOP RATED

Applicant
Gurjeet has applied to or been invited to your or your company's job *Customize IoT Admin Platform/Dashboard*

Proposal Details

$2,200.00
Proposed Bid

Cover letter

What frameworks have you worked with?
ReactJS, VueJS, AngularJS NodeJS Bootstrap

Greetings!

Have gone through the attached image and can work in your timezone.
Sure can assist you in Customize IoT Admin Platform/Dashboard by implementing the changes to the existing platform.

>>Please share the link of GitHub repository.

-Skilled in website design and development , UI/UX, Photoshop, Dashboard
- Well versed with Html5, CSS3, Javascript, Angular, ReactJS, VueJS, Webpack
- Experienced with REST/JSON/SOAP API, 3rd party API,

Some of the projects we have done:
https://gatsby.ai/
https://totaltraining.com/